

Diaphragm moment joint (double collar) - check sheet

Library dmj_lib.m, manual diaphragm_joint_manual.md. Forces from /home/winpatubuntu/scripting/structures/joint_calculations/etabs_joints/gg.txt, joint database built 25-Sep-2026 16:18:20; sheet generated 25-Sep-2026 16:21:37. RSA combinations are excluded. Load combinations matching 'E[xy]' are reported as seismic, the rest as gravity. One section per joint follows.

Summary

Joint	Column	Strong	Weak	Governing check	DCR	Case
10	50	[38 35]	[63 64]	S8 shelf: axial + bending [$M_p(1-n^2)$]	0.78	1.2D+L+1.6Lr
13	53	3	[14 16]	P9 tab: wall vs plate, $t_p \leq F_u t/F_y$	0.94	1.4D

Joint 10

Joint 10, interior (X), 4 beams (h = 160, bf = 82) on HSS 200x100x4, column 50. Strong beams [38 35], weak beams [63 64]. Companion to dmj_lib.m and diaphragm_joint_manual.md. Units N, mm throughout; results shown in kN and kNm. AISC 360-16, LRFD. Class (W N E S): M M M M; shear only on the shelf []; below the collar on a shear tab [].

Governing: S8 shelf: axial + bending [$M_p(1-n^2)$], DCR 0.78, case 1.2D+L+1.6Lr. Gravity DCR = worst over the gravity combinations, seismic DCR = worst over combinations matching 'E[xy]'.

1. Index of limit states

Tag	Limit state	Reference	Capacity	Gravity DCR	Seismic DCR
T1	flange to plate weld (3-sided)	J2.4 (J2-3)	184.8 kN	0.51	0.48
T2	beam flange gross yielding	J4.1 (J4-1)	136.5 kN	0.69	0.65
T3	beam flange rupture at the weld	J4.1(b), D3 U=1.00	182.0 kN	0.52	0.49
T3b	beam flange block shear	J4.3 (J4-5)	315.2 kN	0.30	0.28
T3c	cap plate block shear tear-out	J4.3 (J4-5)	511.2 kN	0.18	0.17
T3d	shelf plate block shear tear-out	J4.3 (J4-5)	511.2 kN	0.18	0.17
T4s	plate yielding at the lap, strong	J4.1 (J4-1)	470.8 kN	0.20	0.19
T5s	plate strips tension, strong [MODEL]	J4.1 (J4-1)	513.0 kN	0.18	0.17
T6s	plate strips compression, strong [MODEL]	E3 (E3-2)	476.1 kN	0.20	0.19
T12s	plate in-plane bending, strong [INFO]	cf. F11.1	12.184 kNm	0.37	0.34
T10s	wall in-plane tension, strong [INFO]	alt. path	180.0 kN	0.52	0.49
T11s	wall stress interaction, strong [INFO]	von Mises	225.0 MPa	0.48	0.47
T4w	plate yielding at the lap, weak	J4.1 (J4-1)	470.8 kN	0.08	0.07
T5w	plate strips tension, weak [MODEL]	J4.1 (J4-1)	513.0 kN	0.08	0.07
T6w	plate strips compression, weak [MODEL]	E3 (E3-2)	503.5 kN	0.08	0.07
T12w	plate in-plane bending, weak [INFO]	cf. F11.1	12.184 kNm	0.16	0.14
T10w	wall in-plane tension, weak [INFO]	alt. path	90.0 kN	0.44	0.39
T11w	wall stress interaction, weak [INFO]	von Mises	225.0 MPa	0.41	0.44
T7s	collar weld to column, strong	J2.4 (J2-3)	244.3 kN	0.02	0.19
T8s	column walls shear, strong	J4.2 (J4-3)	216.0 kN	0.02	0.21
T9s	wall local yielding, strong [MODEL]	cf. J10.2	127.8 kN	0.03	0.36
Tcs	CIDECT DG9 T8.3(2), strong [INFO]	CIDECT 8.6	340.4 kN	0.01	0.13
T7w	collar weld to column, weak	J2.4 (J2-3)	122.2 kN	0.01	0.25
T8w	column walls shear, weak	J4.2 (J4-3)	108.0 kN	0.01	0.28
T9w	wall local yielding, weak [MODEL]	cf. J10.2	90.0 kN	0.02	0.33
Tcw	CIDECT DG9 T8.3(2), weak [INFO]	CIDECT 8.6	340.4 kN	0.00	0.09
S1	beam web local yielding	J10.2 (J10-3)	123.1 kN	0.12	0.09
S2	beam web crippling at the end	J10.3 (J10-5b)	129.0 kN	0.12	0.08
S3	bearing on the shelf plate	J7 (J7-1)	2214.0 kN	0.01	0.00
S10	shelf plate gross shear at root	J4.2 (J4-3)	147.6 kN	0.05	0.04
S4	shelf plate bending	F11.1 (F11-1)	0.664 kNm	0.64	0.44
S8	shelf: axial + bending [$M_p(1-n^2)$]	F11 / plastic	0.544 kNm *	0.78	0.52
S5	shelf to column weld	J2.4 (J2-3)	366.5 kN	0.04	0.03
S6	beam web shear	G2.1 (G2-1)	120.0 kN	0.13	0.09
M1	beam flexure (no flange holes)	F2.1 (F2-1)	27.878 kNm	0.58	0.54
M4	column flexure, strong	F7.1 (F7-1)	34.229 kNm	0.02	0.23
M5	column flexure, weak	F7.1 (F7-1)	21.089 kNm	0.01	0.24

* the capacity depends on the load case; the value shown is at the case governing the envelope, see section 5. DCR above 1.00 in red. Tags marked [MODEL] or [INFO] are engineering models, not code equations.

2. Data

Materials and sections

Item	Symbol	Value	Note
Elastic modulus	E	200000 MPa	
Electrode	FEXX	480 MPa	
Beam yield / tensile	Fy, Fu	250 / 400 MPa	
Beam depth	h	160 mm	
Beam flange width	bf	82 mm	
Beam flange thickness	tf	7.4 mm	
Beam web thickness	tw	5 mm	
Beam root radius	r	0 mm	
Beam plastic modulus	Zx	123900 mm ³	
Column side, strong direction	D	200 mm	HSS 200x100x4
Column side, weak direction	B	100 mm	
Column wall	t	4 mm	nominal
Column yield / tensile	Fy, Fu	250 / 400 MPa	
Column compression capacity	phiPn	not given	from ETABS; enables M6 (H1-1)

Plates and welds

Item	Symbol	Value	How it is obtained
Cap plate thickness	t_cap	12 mm	input
Shelf plate thickness	t_shf	12 mm	input
Plate yield / tensile	Fy, Fu	250 / 400 MPa	input
Lap over each beam flange	L_lap	80 mm	input; also the weld length each side
Erection gap	gap	15 mm	input
Overhang, side with no beam	w_back	30 mm	input
Plate size, strong direction	Wx	390 mm	D + strips [95 95]
Plate size, weak direction	Wy	290 mm	B + strips [95 95]
Strip beside opening, strong / weak	marg	95 / 95 mm	plate_geometry
Whitmore width, strong / weak	b_eff	174.4 / 174.4 mm	plate_geometry
Flange weld leg	leg_fl	5 mm	three-sided
Collar weld leg	leg_col	4 mm	
Collar weld lines, cap / shelf	n_cap, n_shf	1 / 1	1 = one face, 2 = both faces

Seat switches

Switch	Value used	Note
e_react, reaction eccentricity	55 mm	gap + L_lap/2 (e_auto)
share, part of V on the shelf	0.50	1.0 shelf alone, 0.5 both plates
bending width (wmode)	bf, no dispersion	same width for S4 and S8
ribs under the shelf	none	

3. Demands

Quantity	How it is obtained	Gravity	Seismic
Load combination	from ETABS	1.2D+L+1.6Lr	1.2D+Ey+-.3Ex+L+.2S Max
Beam 38 moment (strong)	M3 at the beam end	16.22 kNm	8.45 kNm
Beam 35 moment (strong)	M3 at the beam end	15.21 kNm	15.16 kNm
Beam 63 moment (weak)	M3 at the beam end	6.88 kNm	2.28 kNm
Beam 64 moment (weak)	M3 at the beam end	6.61 kNm	3.88 kNm
Beam 38 axial	P at the beam end	-0.03 kN	0.41 kN
Beam 35 axial	P at the beam end	0.08 kN	-0.33 kN
Beam 63 axial	P at the beam end	-0.05 kN	0.28 kN
Beam 64 axial	P at the beam end	-0.02 kN	0.07 kN
Column-top moment, strong	M3 at the column top	0.54 kNm	6.78 kNm
Column-top moment, weak	other axis	0.25 kNm	1.55 kNm
Largest reaction	-F1 on the joint, column axes	15.37 kN	10.60 kN
Column axial	-P at the column top	41.63 kN	23.21 kN
Flange force, strong	M /z + N /2, demand of T4s	94.3 kN	88.3 kN
Flange force, weak	M /z + N /2, demand of T4w	40.0 kN	22.6 kN
Net force, strong	M_col / z, demand of T7s	3.1 kN	39.4 kN
Net force, weak	M_col / z, demand of T7w	1.5 kN	9.0 kN

The gravity and seismic columns are the combinations with the largest DCR of any check in each group.

4. Capacities, with numbers substituted

Tag	Reference	Substitution	Result
T1	J2.4 (J2-3)	-	184.8 kN
T2	J4.1 (J4-1)	-	136.5 kN
T3	J4.1(b), D3 U=1.00	-	182.0 kN
T3b	J4.3 (J4-5)	-	315.2 kN
T3c	J4.3 (J4-5)	-	511.2 kN
T3d	J4.3 (J4-5)	-	511.2 kN
T4s	J4.1 (J4-1)	-	470.8 kN
T5s	J4.1 (J4-1)	-	513.0 kN
T6s	E3 (E3-2)	-	476.1 kN
T12s	cf. F11.1	-	12.184 kNm
T10s	alt. path	-	180.0 kN
T11s	von Mises	-	225.0 MPa
T4w	J4.1 (J4-1)	-	470.8 kN
T5w	J4.1 (J4-1)	-	513.0 kN
T6w	E3 (E3-2)	-	503.5 kN
T12w	cf. F11.1	-	12.184 kNm
T10w	alt. path	-	90.0 kN
T11w	von Mises	-	225.0 MPa
T7s	J2.4 (J2-3)	-	244.3 kN
T8s	J4.2 (J4-3)	-	216.0 kN
T9s	cf. J10.2	-	127.8 kN
Tcs	CIDECT 8.6	-	340.4 kN
T7w	J2.4 (J2-3)	-	122.2 kN
T8w	J4.2 (J4-3)	-	108.0 kN
T9w	cf. J10.2	-	90.0 kN
Tcw	CIDECT 8.6	-	340.4 kN
S1	J10.2 (J10-3)	-	123.1 kN
S2	J10.3 (J10-5b)	-	129.0 kN
S3	J7 (J7-1)	-	2214.0 kN
S10	J4.2 (J4-3)	-	147.6 kN
S4	F11.1 (F11-1)	-	0.664 kNm
S8	F11 / plastic	-	0.544 kNm
S5	J2.4 (J2-3)	-	366.5 kN
S6	G2.1 (G2-1)	-	120.0 kN
M1	F2.1 (F2-1)	-	27.878 kNm
M4	F7.1 (F7-1)	-	34.229 kNm
M5	F7.1 (F7-1)	-	21.089 kNm

dmj_lib.m does not supply the substituted formulas; the table shows the capacities only.

5. Capacities that depend on the load case

These capacities change with the forces of the case (for example the axial force in the shelf, or the column stress through Qf). Each is shown at its worst gravity and worst seismic combination.

Tag	Limit state	Gravity case	Capacity	Demand	DCR	Seismic case	Capacity	Demand	DCR
S8	shelf: axial + bending [$M_p(1-n^2)$]	1.2D+L+1.6Lr	0.544 kNm	0.423 kNm	0.78	1.2D+Ey+-.3Ex+ L+.2S Max	0.559 kNm	0.291 kNm	0.52

6. Checks to do first

1. S8 shelf: axial + bending [$M_p(1-n^2)$] (F11 / plastic): 0.544 kNm, against a demand of 0.423 kNm in 1.2D+L+1.6Lr.
2. T2 beam flange gross yielding (J4.1 (J4-1)): 136.5 kN, against a demand of 94.3 kN in 1.2D+L+1.6Lr.
3. T8s column walls shear, strong (J4.2 (J4-3)): 216.0 kN, against a demand of 45.8 kN in 1.3D+-Ey+.3Ex+L+.2S Max.

If these match, the units, the resistance factors and the forces are right, and the rest is the same arithmetic with different areas.

7. What is not on this sheet

Column P-M-M interaction unless ϕP_n is given (take it from ETABS), beam torsion, lateral-torsional buckling of the cantilevers, the erection cleat, fatigue, fire, and seismic detailing under AISC 341 (this connection is not prequalified under AISC 358). Checks marked [MODEL] or [INFO] carry no code equation: T5, T6, T9, T10, T11, Tc and the lever arm z. Section 8 of the manual lists them with their risk.

Joint 13

Joint 13, T, 3 beams (h = 160, bf = 82) on HSS 200x100x4, column 53. Strong beams 3, weak beams [14 16]. Companion to dmj_lib.m and diaphragm_joint_manual.md. Units N, mm throughout; results shown in kN and kNm. AISC 360-16, LRFD. Class (W N E S): M M M NL; shear only on the shelf []; below the collar on a shear tab 70.

Governing: P9 tab: wall vs plate, $t_p \leq F_u t / F_y$, DCR 0.94, case 1.4D. Gravity DCR = worst over the gravity combinations, seismic DCR = worst over combinations matching 'E[xy]'.

1. Index of limit states

Tag	Limit state	Reference	Capacity	Gravity DCR	Seismic DCR
T1	flange to plate weld (3-sided)	J2.4 (J2-3)	184.8 kN	0.15	0.33
T2	beam flange gross yielding	J4.1 (J4-1)	136.5 kN	0.21	0.44
T3	beam flange rupture at the weld	J4.1(b), D3 U=1.00	182.0 kN	0.16	0.33
T3b	beam flange block shear	J4.3 (J4-5)	315.2 kN	0.09	0.19
T3c	cap plate block shear tear-out	J4.3 (J4-5)	511.2 kN	0.06	0.12
T3d	shelf plate block shear tear-out	J4.3 (J4-5)	511.2 kN	0.06	0.12
T4s	plate yielding at the lap, strong	J4.1 (J4-1)	470.8 kN	0.02	0.13
T5s	plate strips tension, strong [MODEL]	J4.1 (J4-1)	513.0 kN	0.02	0.12
T6s	plate strips compression, strong [MODEL]	E3 (E3-2)	476.1 kN	0.02	0.13
T12s	plate in-plane bending, strong [INFO]	cf. F11.1	12.184 kNm	0.04	0.24
T10s	wall in-plane tension, strong [INFO]	alt. path	180.0 kN	0.06	0.33
T11s	wall stress interaction, strong [INFO]	von Mises	225.0 MPa	0.12	0.47
T4w	plate yielding at the lap, weak	J4.1 (J4-1)	470.8 kN	0.06	0.08
T5w	plate strips tension, weak [MODEL]	J4.1 (J4-1)	513.0 kN	0.06	0.08
T6w	plate strips compression, weak [MODEL]	E3 (E3-2)	503.5 kN	0.06	0.08
T12w	plate in-plane bending, weak [INFO]	cf. F11.1	12.184 kNm	0.11	0.15
T10w	wall in-plane tension, weak [INFO]	alt. path	90.0 kN	0.32	0.43
T11w	wall stress interaction, weak [INFO]	von Mises	225.0 MPa	0.29	0.45
T7s	collar weld to column, strong	J2.4 (J2-3)	244.3 kN	0.05	0.25
T8s	column walls shear, strong	J4.2 (J4-3)	216.0 kN	0.06	0.28
T9s	wall local yielding, strong [MODEL]	cf. J10.2	127.8 kN	0.10	0.47
Tcs	CIDECT DG9 T8.3(2), strong [INFO]	CIDECT 8.6	340.4 kN	0.04	0.18
T7w	collar weld to column, weak	J2.4 (J2-3)	122.2 kN	0.03	0.21
T8w	column walls shear, weak	J4.2 (J4-3)	108.0 kN	0.03	0.23
T9w	wall local yielding, weak [MODEL]	cf. J10.2	90.0 kN	0.04	0.28
Tcw	CIDECT DG9 T8.3(2), weak [INFO]	CIDECT 8.6	340.4 kN	0.01	0.07
S1	beam web local yielding	J10.2 (J10-3)	123.1 kN	0.05	0.06
S2	beam web crippling at the end	J10.3 (J10-5b)	129.0 kN	0.05	0.06
S3	bearing on the shelf plate	J7 (J7-1)	2214.0 kN	0.00	0.00
S10	shelf plate gross shear at root	J4.2 (J4-3)	147.6 kN	0.02	0.02
S4	shelf plate bending	F11.1 (F11-1)	0.664 kNm	0.26	0.30
S8	shelf: axial + bending [$M_p(1-n^2)$]	F11 / plastic	0.615 kNm *	0.26	0.33
S5	shelf to column weld	J2.4 (J2-3)	366.5 kN	0.02	0.02
S6	beam web shear	G2.1 (G2-1)	120.0 kN	0.05	0.06
P1	tab: bolt shear	J3.6 (J3-1)	112.2 kN	0.01	0.01
P2	tab: bolt bearing on the plate	J3.10 (J3-6)	103.7 kN	0.01	0.00
P3	tab: bolt bearing on the beam web	J3.10 (J3-6)	115.2 kN	0.01	0.00
P4	tab: web tear-out to the beam end (N)	J3.10 (J3-6c)	75.6 kN	0.01	0.01
P5	tab: plate shear yielding	J4.2 (J4-3)	90.0 kN	0.01	0.01
P6	tab: plate shear rupture	J4.2 (J4-4)	69.1 kN	0.02	0.01
P7	tab: plate block shear	J4.3 (J4-5)	88.4 kN	0.01	0.01
P8	tab: welds to the column wall (V, V ea, N)	J2.4 (J2-3)	122.2 kN	0.03	0.01
P9	tab: wall vs plate, $t_p \leq F_u t / F_y$	K2.2 long. plate shear	6.40 mm	0.94	0.94
P10	tab: beam web net shear rupture	J4.2 (J4-4)	111.6 kN	0.01	0.00
P11	tab: face B/t ≤ 40 , K2.2A [INFO]	K2.2A	40.000	0.62	0.62
M1	beam flexure (no flange holes)	F2.1 (F2-1)	27.878 kNm	0.18	0.37
M4	column flexure, strong	F7.1 (F7-1)	34.229 kNm	0.07	0.30
M5	column flexure, weak	F7.1 (F7-1)	21.089 kNm	0.03	0.21
S7	flange weld under uplift	J2.4 (J2-3)	184.8 kN	-	0.02

* the capacity depends on the load case; the value shown is at the case governing the envelope, see section 5. DCR above 1.00 in red. Tags marked [MODEL] or [INFO] are engineering models, not code equations.

2. Data

Materials and sections

Item	Symbol	Value	Note
Elastic modulus	E	200000 MPa	
Electrode	FEXX	480 MPa	
Beam yield / tensile	Fy, Fu	250 / 400 MPa	
Beam depth	h	160 mm	
Beam flange width	bf	82 mm	
Beam flange thickness	tf	7.4 mm	
Beam web thickness	tw	5 mm	
Beam root radius	r	0 mm	
Beam plastic modulus	Zx	123900 mm ³	
Column side, strong direction	D	200 mm	HSS 200x100x4
Column side, weak direction	B	100 mm	
Column wall	t	4 mm	nominal
Column yield / tensile	Fy, Fu	250 / 400 MPa	
Column compression capacity	phiPn	not given	from ETABS; enables M6 (H1-1)

Plates and welds

Item	Symbol	Value	How it is obtained
Cap plate thickness	t_cap	12 mm	input
Shelf plate thickness	t_shf	12 mm	input
Plate yield / tensile	Fy, Fu	250 / 400 MPa	input
Lap over each beam flange	L_lap	80 mm	input; also the weld length each side
Erection gap	gap	15 mm	input
Overhang, side with no beam	w_back	30 mm	input
Plate size, strong direction	Wx	390 mm	D + strips [95 95]
Plate size, weak direction	Wy	290 mm	B + strips [95 95]
Strip beside opening, strong / weak	marg	95 / 95 mm	plate_geometry
Whitmore width, strong / weak	b_eff	174.4 / 174.4 mm	plate_geometry
Flange weld leg	leg_fl	5 mm	three-sided
Collar weld leg	leg_col	4 mm	
Collar weld lines, cap / shelf	n_cap, n_shf	1 / 1	1 = one face, 2 = both faces

Seat switches

Switch	Value used	Note
e_react, reaction eccentricity	55 mm	gap + L_lap/2 (e_auto)
share, part of V on the shelf	0.50	1.0 shelf alone, 0.5 both plates
bending width (wmode)	bf, no dispersion	same width for S4 and S8
ribs under the shelf	none	

3. Demands

Quantity	How it is obtained	Gravity	Seismic
Load combination	from ETABS	1.4D	1.2D+Ex+.3Ey+L+.2S Max
Beam 3 moment (strong)	M3 at the beam end	0.90 kNm	1.69 kNm
Beam 14 moment (weak)	M3 at the beam end	2.47 kNm	6.71 kNm
Beam 16 moment (weak)	M3 at the beam end	2.38 kNm	2.28 kNm
Beam 3 axial	P at the beam end	-0.01 kN	0.01 kN
Beam 14 axial	P at the beam end	-0.04 kN	-0.18 kN
Beam 16 axial	P at the beam end	-0.03 kN	0.20 kN
Column-top moment, strong	M3 at the column top	1.02 kNm	1.55 kNm
Column-top moment, weak	other axis	0.12 kNm	4.32 kNm
Largest reaction	-F1 on the joint, column axes	2.93 kN	4.85 kN
Column axial	-P at the column top	8.12 kN	8.50 kN
Flange force, strong	M /z + N /2, demand of T4s	5.2 kN	9.8 kN
Flange force, weak	M /z + N /2, demand of T4w	14.4 kN	39.1 kN
Net force, strong	M_col / z, demand of T7s	5.9 kN	9.0 kN
Net force, weak	M_col / z, demand of T7w	0.7 kN	25.1 kN

The gravity and seismic columns are the combinations with the largest DCR of any check in each group.

4. Capacities, with numbers substituted

Tag	Reference	Substitution	Result
T1	J2.4 (J2-3)	-	184.8 kN
T2	J4.1 (J4-1)	-	136.5 kN
T3	J4.1(b), D3 U=1.00	-	182.0 kN
T3b	J4.3 (J4-5)	-	315.2 kN
T3c	J4.3 (J4-5)	-	511.2 kN
T3d	J4.3 (J4-5)	-	511.2 kN
T4s	J4.1 (J4-1)	-	470.8 kN
T5s	J4.1 (J4-1)	-	513.0 kN
T6s	E3 (E3-2)	-	476.1 kN
T12s	cf. F11.1	-	12.184 kNm
T10s	alt. path	-	180.0 kN
T11s	von Mises	-	225.0 MPa
T4w	J4.1 (J4-1)	-	470.8 kN
T5w	J4.1 (J4-1)	-	513.0 kN
T6w	E3 (E3-2)	-	503.5 kN
T12w	cf. F11.1	-	12.184 kNm
T10w	alt. path	-	90.0 kN
T11w	von Mises	-	225.0 MPa
T7s	J2.4 (J2-3)	-	244.3 kN
T8s	J4.2 (J4-3)	-	216.0 kN
T9s	cf. J10.2	-	127.8 kN
Tcs	CIDECT 8.6	-	340.4 kN
T7w	J2.4 (J2-3)	-	122.2 kN
T8w	J4.2 (J4-3)	-	108.0 kN
T9w	cf. J10.2	-	90.0 kN
Tcw	CIDECT 8.6	-	340.4 kN
S1	J10.2 (J10-3)	-	123.1 kN
S2	J10.3 (J10-5b)	-	129.0 kN
S3	J7 (J7-1)	-	2214.0 kN
S10	J4.2 (J4-3)	-	147.6 kN
S4	F11.1 (F11-1)	-	0.664 kNm
S8	F11 / plastic	-	0.615 kNm
S5	J2.4 (J2-3)	-	366.5 kN
S6	G2.1 (G2-1)	-	120.0 kN
P1	J3.6 (J3-1)	-	112.2 kN
P2	J3.10 (J3-6)	-	103.7 kN
P3	J3.10 (J3-6)	-	115.2 kN
P4	J3.10 (J3-6c)	-	75.6 kN
P5	J4.2 (J4-3)	-	90.0 kN
P6	J4.2 (J4-4)	-	69.1 kN

Tag	Reference	Substitution	Result
P7	J4.3 (J4-5)	-	88.4 kN
P8	J2.4 (J2-3)	-	122.2 kN
P9	K2.2 long. plate shear	-	6.40 mm
P10	J4.2 (J4-4)	-	111.6 kN
P11	K2.2A	-	40.000
M1	F2.1 (F2-1)	-	27.878 kNm
M4	F7.1 (F7-1)	-	34.229 kNm
M5	F7.1 (F7-1)	-	21.089 kNm
S7	J2.4 (J2-3)	-	184.8 kN

dmj_lib.m does not supply the substituted formulas; the table shows the capacities only.

5. Capacities that depend on the load case

These capacities change with the forces of the case (for example the axial force in the shelf, or the column stress through Qf). Each is shown at its worst gravity and worst seismic combination.

Tag	Limit state	Gravity case	Capacity	Demand	DCR	Seismic case	Capacity	Demand	DCR
S8	shelf: axial + bending [Mp(1-n^2)]	1.2D+L+1.6Lr	0.653 kNm	0.172 kNm	0.26	1.3D+-Ey+.3Ex+ L+.2S Max	0.615 kNm	0.200 kNm	0.33

6. Checks to do first

1. P9 tab: wall vs plate, $t_p \leq F_u t / F_y$ (K2.2 long. plate shear): 6.40 mm, against a demand of 6.00 mm in 1.4D.
2. T2 beam flange gross yielding (J4.1 (J4-1)): 136.5 kN, against a demand of 60.3 kN in 1.3D+-Ey+.3Ex+L+.2S Max.
3. T8s column walls shear, strong (J4.2 (J4-3)): 216.0 kN, against a demand of 60.2 kN in 1.3D+-Ey+.3Ex+L+.2S Max.

If these match, the units, the resistance factors and the forces are right, and the rest is the same arithmetic with different areas.

7. What is not on this sheet

Column P-M-M interaction unless ϕP_n is given (take it from ETABS), beam torsion, lateral-torsional buckling of the cantilevers, the erection cleat, fatigue, fire, and seismic detailing under AISC 341 (this connection is not prequalified under AISC 358). Checks marked [MODEL] or [INFO] carry no code equation: T5, T6, T9, T10, T11, Tc and the lever arm z. Section 8 of the manual lists them with their risk.